

BDS DSC312 - Computer Vision with Multi-modal models & Analytics

Project Phase 1: Problem Definition, Literature Survey, and Proof of Concept (PoC)

Overview

It's time to turn your proposals into reality! We are officially moving into Stage 1 of the Computer Vision Project. You will convert your high-level idea into a technical roadmap. You must prove that your problem is grounded in existing research, that your code "runs", and that your data pipeline is "live". A Proof of Concept (PoC) is not a finished product; it's a feasibility test. I want you to have sufficient evidence that you can complete your project on time. In order to do that, you need to be able to do three things:

- 1- **Data Access** and EDA (Exploratory Data Analysis): Do you have the actual images on your machine/cloud storage? Can you load and visualise a few of them? Did you check for class imbalance? image sizes? Can you resize, normalise, and augment the dataset?
- 2- **Pipeline setup**: Can you load at least a subset of the dataset, pass it through a basic pre-trained model (even ResNet-18), and get *any* output?
- 3- **Baseline code**: What happened when you ran the baseline? Did it crash? Unacceptable! Observe the loss curves and note the starting accuracy, even if it is 10%. That's OK. We just need a starting point and a plan on how to improve it.

Document all your findings in a **report** that also includes a mini literature survey as described.

Logistics

Total Marks: 30% of the unit's total.

Deliverables: (**one submission** only per team)

- Jupyter Notebook (.ipynb)
- Report:
 - Format: PDF file.
 - Word count: Strictly enforced 1250 words max. **We will stop reading past the count.**
 - Page limit: 2-3 pages maximum (excludes references and figures).
 - Use the provided IEEE Conference Template.

Timeline:

- Team Selection: By Monday, 9/3/2026 - 5 PM
- Stage 1 Completion: Thu 19/3/2026 - 11:59 PM. (1 day late -10%, 2 days late -25%, no later).

1. The Teams Spreadsheet

https://spjainorg-my.sharepoint.com/:x:/g/personal/hazem_elalfy_spjain_edu_au/IQCGmQU3taXtSo8lqKM7z9hSAQbEoz8yx9JEFiZlpYxvVg?e=jnRtk7

- You must finalise your groups in the [shared spreadsheet](#) by Mon 9/3 (5 PM).
 - Maximum **group size 4** students. (One group will be 3 students).
 - Copy your name from the list on the right. Those are the students who submitted a proposal.
 - Do **not** add students who are not in the right-hand side list to teams.
 - I created two sample groups based on your proposals. Follow the example I created.
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2. Report Structure

We will strictly enforce the word counts below. **We will stop reading once the limit is reached.** Use the IEEE two-column template to maintain a professional standard. Each column is about 230-250 words.

Section I: Problem Definition (max 250 words)

Describe the task. What is the input, what is the output, and what is challenging about it?

Section II: Mini Literature Survey (max 500 words)

A literature survey summarises what the scientific community has already tried. You are required to locate your project within the current body of work.

- Why? Don't "reinvent the wheel". Instead, identify where you can improve.
- How? Use Google Scholar. Look for papers from the last 5 years.
- Requirement: Cite and describe at least **5 relevant papers** or established architectures (e.g., "Our approach builds on the Feature Pyramid logic introduced by Lin et al. (2017)...").

Section III: Exploratory Data Analysis (max 250 words)

Describe your dataset. Why did you choose it? (other than because I told you to do so). Include sample images (from different classes) and a brief note on class distribution. Is the data "clean," or is there a "Silent Killer" (e.g., 251976 class imbalance)? Will you preprocess it? How?

Section IV: The Baseline and Proof of Concept (max 250 words)

Describe your initial experiment. Which pre-trained backbone did you use? Why is this backbone suitable for your problem? What was the result of your first "Forward Pass"?

- Include a **plot** of your initial training/validation loss curves or sample prediction outputs.
- *Note:* If your project does not use a standard loss curve (e.g., a tracking project), provide a screenshot of the output detection boxes or feature embeddings instead to prove the pipeline is functional.

Section V: References (no word limit)

It is no secret that everyone uses AI. If that is your first time doing a literature review, note that ChatGPT and Copilot hallucinate non-existent references. Perplexity and Gemini are better in that aspect. But most AI cite irrelevant references. If I find such blind use of AI, I will give zero for the literature survey section.

3. The Proof-of-Concept Code:

Submit a single Jupyter Notebook.

- It must be organised with Markdown headers. Notebooks with only code cells and no Markdown documentation will be penalised under the 'Professionalism' marking rubric.
 - It must show the code used to load data and run the baseline model.
 - I reserve the right to randomly select and execute any student's code to verify that the results match the plots/screenshots provided in the report. Ensure your paths are relative and your code is reproducible.
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4. Marking Rubric (30 Marks)

Rubric	Marks	Criteria
Problem definition	5	Clear problem definition and explanation of challenge.
Literature survey	5	5+ relevant and recent academic citations.
Dataset and EDA	5	Evidence of successfully loading the data and running a few training epochs.
Baseline	5	Report: Describe the architecture, metrics, and initial results (plots/screenshots).
Proof of Concept Code	5	Code: organised Notebook. Does not crash. Produces results similar to those documented.
Writing structure	5	Well-structured, clear and effective technical writing. Adherence to the IEEE template and strict word counts.

5. Lookahead...

In Stage 2, we will have a "Best Project" recognition for teams that outperform published results or provide an exceptionally creative solution. If you want to reach there, this will require an adequate literature review in Stage 1.

Good luck. Let's build something impressive!

- Hazem El-Alfy